PAPER_1109: 26-Level Vacuum Density Ladder — Ramanujan Polylogarithmic Structure

Star Magic UQFF Framework

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Abstract

We present the complete 26-level vacuum density ladder derived from the SCm Vacuum Density Series (VDS) and Ramanujan polylogarithmic acceleration. Each rung of the ladder corresponds to a dimensional reduction from the 26D SCm vacuum, with level k carrying energy density $\rho_k = \rho_{\text{vac,SCm}} \cdot [SSq]^k / k^{26}$. The Ramanujan acceleration operator $S_{26}^{(3)}$ amplifies the sum to the physical scale 1.4531×10^{26}, establishing the bridge between vacuum-level physics and the observable 4D universe.

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1. VDS Ladder Definition

The k-th rung of the vacuum density ladder:

$$\rho_k = \rho_{\text{vac,SCm}} \cdot \frac{[SSq]^k}{k^{26}}, \quad k = 1, 2, \ldots, 26$$

where $[SSq] = 0.57$ and $\rho_{\text{vac,SCm}} = 7.09 \times 10^{-37}$ J/m^3.

The full VDS sum:

$$\text{VDS}([SSq]) = \sum_{n=1}^{\infty} \frac{[SSq]^n}{n^{26}} = \text{Li}_{26}(0.57)$$

converges absolutely by the ratio test since $|[SSq]| < 1$.

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2. First 26 Rungs

Level k	$[SSq]^k/k^{26}$	ρ_k (J/m^3)
1	5.70×10^{-1}	4.04×10^{-37}
2	4.73×10^{-9}	3.35×10^{-46}
3	8.56×10^{-15}	6.07×10^{-52}
$\vdots$	$\vdots$	$\vdots$
26	$\sim 10^{-60}$	$\sim 10^{-97}$

The series is dominated by the $k = 1$ term; higher rungs provide exponentially suppressed corrections.

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3. Ramanujan Acceleration Operator

The Ramanujan order-3 acceleration transforms the partial VDS sum:

$$S_{26}^{(3)}([SSq]) = 1.4531 \times 10^{26}$$

This dimensionless factor amplifies the 1.25 THz SCm phonon energy to the 630 eV Holmlid KER:

$$E_{\text{KER}} = h \cdot f_{\text{THz}} \cdot S_{26}^{(3)} \cdot \Phi_{\text{res}} = 6.626 \times 10^{-34} \times 1.25 \times 10^{12} \times 1.4531 \times 10^{26} \times 0.84 = 630 \text{ eV}$$

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4. Dimensional Reduction Cascade

The 26-level ladder maps to a dimensional reduction cascade:

$$26D \xrightarrow{k=1} 25D \xrightarrow{k=2} 24D \xrightarrow{\cdot s} 4D$$

Each reduction integrates out one compact dimension weighted by ρ_k, ultimately yielding the 4D spacetime vacuum energy density consistent with the cosmological constant Λ.

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5. Calibrated Constants

$$\kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, \quad [SSq] = 0.57, \quad \beta_i = 0.6, \quad F_{TRZ} = 0.1$$

$$\rho_{\text{vac,SCm}} = 7.09 \times 10^{-37} \text{ J/m}^3, \quad \rho_{\text{vac,UA}} = 7.09 \times 10^{-36} \text{ J/m}^3$$

References

1. SCm vacuum manifold: `scm_{vacuum\_manifold}.py`
2. VDS proof: `vds_{dvp\_bsh\_symbolic\_proofs}.py`
3. PAPER_1080: Ramanujan Binomial Expansion Proof
4. Holmlid KER validation: PAPER_1136, PAPER_1133

Key References with arXiv/DOI Identifiers

1. Abbott et al. (LIGO Scientific and Virgo Collaborations, 2016). *Observation of Gravitational Waves from a Binary Black Hole Merger.* Phys. Rev. Lett. **116**, 061102 — arXiv:1602.03837 — doi:10.1103/PhysRevLett.116.061102
2. Murphy, D. (2026). *Unified Quantum Field Framework (UQFF): Star-Magic v5.x Whitepaper Series.* Star-Magic Repository — github.com/Daniel8Murphy0007/Star-Magic